FORMATIVE ASSESSMENT – 4 (FA-4)

MATHEMATICS • CLASS 9 • GENERATING NEW PROBLEMS

Problem 1

The four angles of a quadrilateral are in the continuous ratio **3 : 5 : 9 : 13**. Find the numerical measure of each angle. Formulate a problem where three angles are known and the fourth angle must be calculated.

Problem 2

Show that the diagonals of a rhombus are mutually perpendicular to each other. Write a problem using this property to determine the side length of a rhombus when its two diagonal lengths are given as **12 cm** and **16 cm**.

Problem 3

A chord of length **16 cm** is at a perpendicular distance of **6 cm** from the centre of a circle. Calculate the radius of the circle and formulate a similar problem to determine the distance when radius and chord length are given.

Problem 4

In a cyclic quadrilateral **ABCD**, the opposite angles satisfy $\angle A = (2x + 4)^\circ$ and $\angle C = (3x - 24)^\circ$. Find the value of **x** and compute all four interior angles if $\angle B = (y + 10)^\circ$ and $\angle D = (2y - 40)^\circ$.

Problem 5

The mean of 5 observations is **18**. If one observation is excluded, their mean becomes **16**. Find the excluded value. Formulate an extension where an additional observation is included instead.